

Systems Design Engineering Portfolio

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Mar 2026

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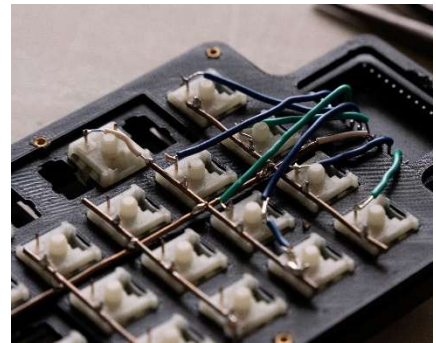


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Media Creation and Capturing

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<https://www.instagram.com/netthenut/>

Audio Spectrum Visualizer Desktop Device

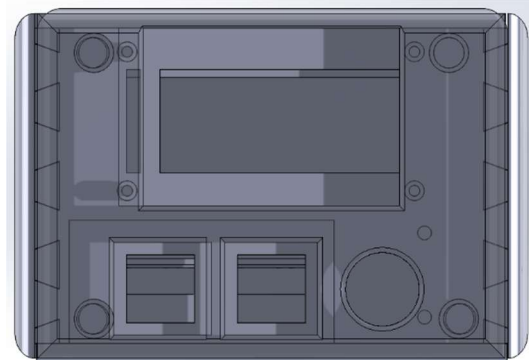
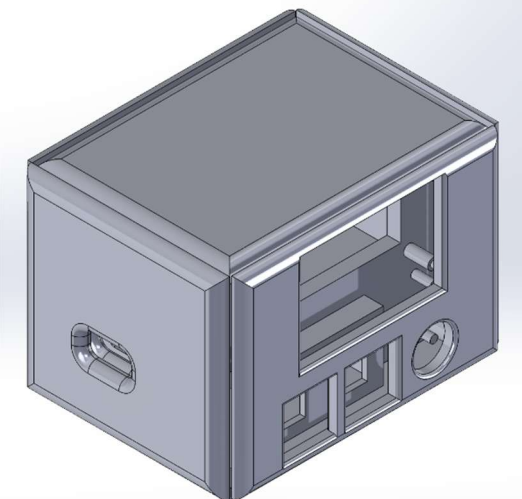
<https://www.youtube.com/watch?v=QmaiqrSgZdo>

- A desktop device which takes Windows media playback (music, video, movie), takes the sound data and media information from the media and displays it on the screen.
- Primarily 3d printed (as it's supposed to be replicable and user-modular) using PLA for all structural components with PETG and copper foil tape for the 3D printed PCB. (**Bambu Labs P1S**)
- Bottom / Top mount made of wood, milled using the **Roland MDX-40 CNC Mill**. Done using a 6mm SE and 1/8 BE.
- Modelled in **SolidWorks**, coded in **C++** (Arduino IDE) and **Python** 3.11.9.



Results

- A device that performs spectral analysis on Windows media playback, sends band audio data to an ESP32 controller via serial communications, and displays it on a 2-inch TFT display along with media metadata.
- The device can change Windows volume, play/pause and skip/back media with the buttons and rotary encoder via quadrature decoding and filtering for the rotary encoder, and serial communications.
- Magnetic side panels in combination with dovetail joints for the main frame create a modular, sturdy, and satisfying product to handle and use.
- In a theoretical finished product, the main frame, electronics, and a base set of magnetic shrouds would



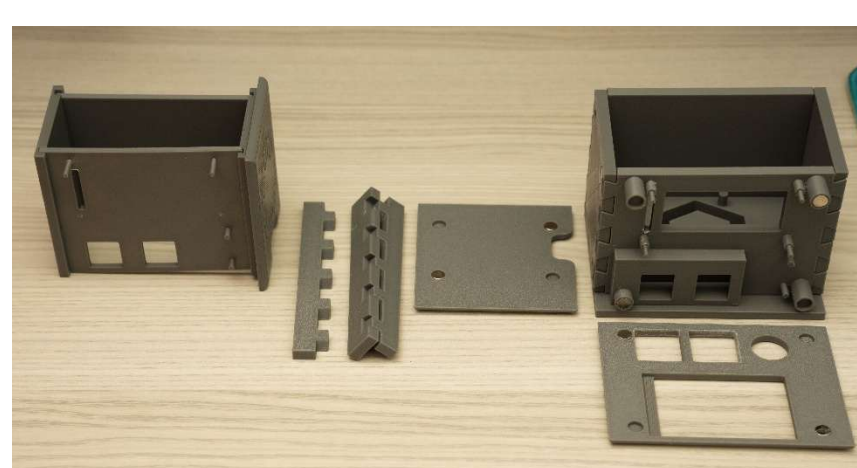
be included with the product kit. CAD files for the magnetic shrouds would be included for users to create and customize their product while also opening allowing CNC wood or metal fabrication.

Next Steps

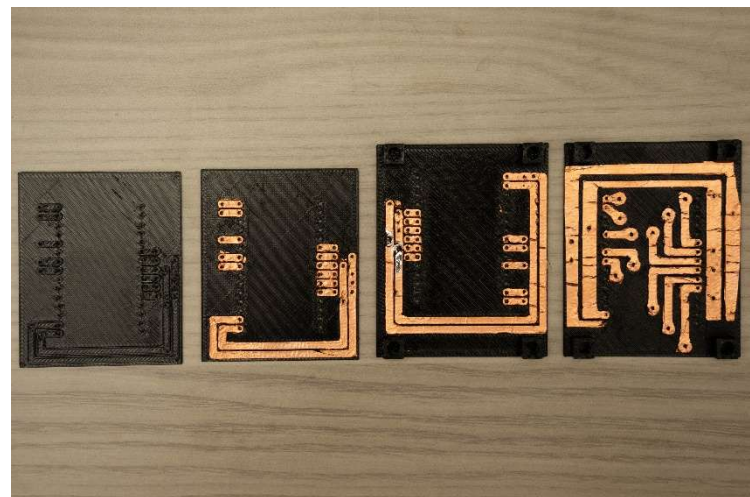
- Create the product trailer video with media captured from the design and production process and create a suite of mock advertising media for the product. (Youtube video with process work attached above)
- Explore manufacturing methods by creating a wooden knob, keycaps, and shrouds as replacements for the 3d printed components. (Epilog Fusion M2 Laser Cutter or more CNC milled components)



Early Prototype of the device. All electrical components were fully functional.



Version 1 pictured on the left, Dovetail join tests in the middle, and Version 2 pictured on the right



All failed version of the 3D Printed + copper foil tape PCB.

Printable Arc Notebook

- A reverse-engineered system to emulate the existing *Staples Arc Notebook* system, a modular binder/notebook system
- Designed to be fully 3d printable which also heavily restricted material use and form
- The main notebook components were engineered to fit with already existing paper holepunch tolerances
- A mix of materials and properties were explored in this project (PLA, TPU95A)

Results

- A working emulation that is easily printable, assemble-able, and opens an entire ecosystem of printable accessories
- Iterative designs to approach a design that is polished, works like the native system, and feels satisfying to use
 1. Early versions did not feel good to use because of the thickness and geometry of the covers, the carry size, and "flipability" were concerns)
- Wear testing was done on the circular spine rings and the covers to test the feasibility of the final design printed in one singular material
 1. Repeated insertions of the cover to the circular spines as well as bending the cover just to the point of plastic deformation (figures
- Iterations included remeasuring and reprinting to get tolerances correct and snug
- Experimentation of material properties (PLA and TPU95A), as early versions of the cover included a TPU95A spine



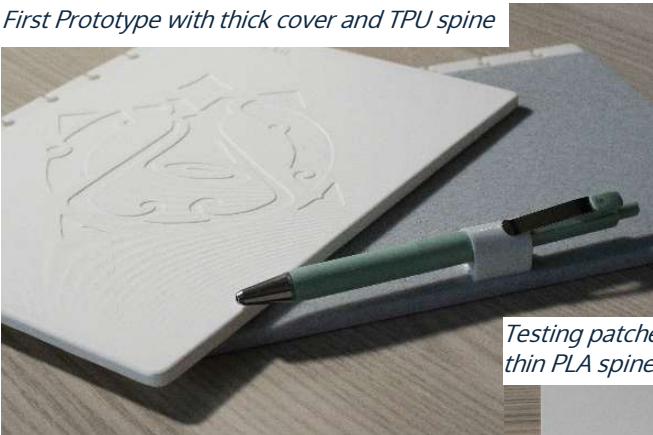
Final prototype onto of previous prototypes



First Prototype with thick cover and TPU spine



First Prototype with thick cover and TPU spine



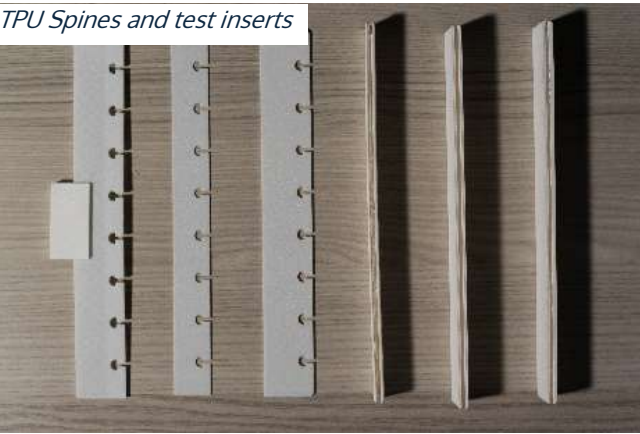
Testing patches for the thin PLA spine



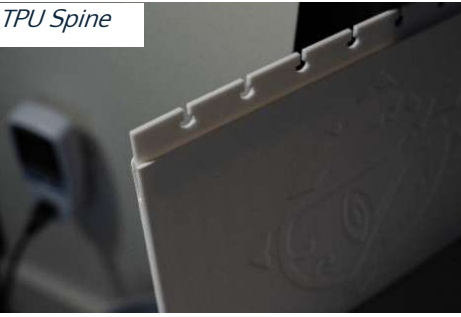
Testing patches for the thin PLA spine



TPU Spines and test inserts



TPU Spine



TPU Pen holder variations for tolerances



Belt Mounted Monopod Support

- A photo/videography accessory to replace the purchase of a specialized vest
- Designed with load considerations, and usability / human metrics
- Contained multiple iterations/prototypes and a final assembly that contains two parts: the body mount and the monopod mount

Results

- A successful design that will be used on future photoshoots as it greatly increases stability and reduces strain on my neck, shoulders, and back.
- **Successful and meaningful changes in between prototypes:**
 1. Version 1 was not large enough for the foot of the monopod. It did not resist the twisting moment of the camera assembly enough. Generally inconvenient mounting system. (*figures 1-3*)
 2. Version two was dimensioned properly and included a "clip" system instead of the slot. Still did not correctly resist the twisting moment. (*figures 4-6*)
 3. Version three had increased size to spread the load across the belt and the body. Still twisted when force was applied. (*figures 7-9*)
 4. Version 3.5 and 4 included a separate larger body mount to distribute the load across a large surface of the body, while also including a larger section of the belt. The Monopod foot cup was also slightly changed to help when the monopod is fully horizontal (*figures 10-13*)



Early planning image

Next Steps

- There aren't necessarily structural concerns, but PETG could be a good fit due to the increased UV resistance and easy printability

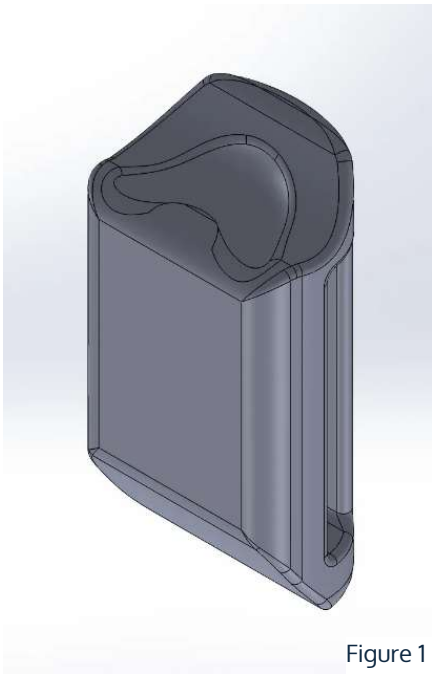


Figure 1

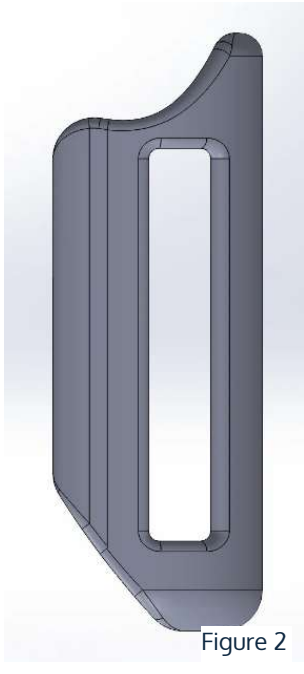


Figure 2



Figure 3

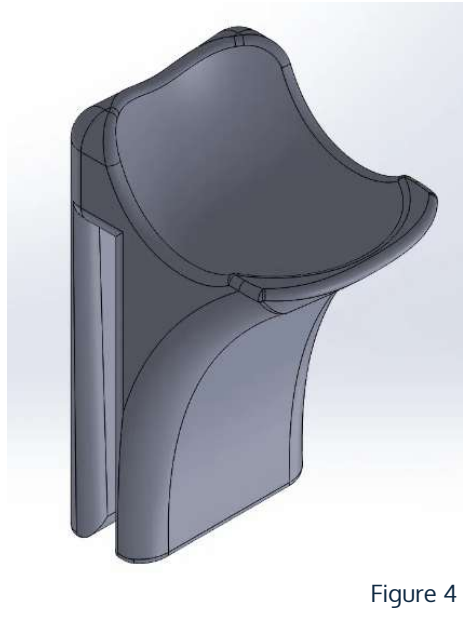


Figure 4

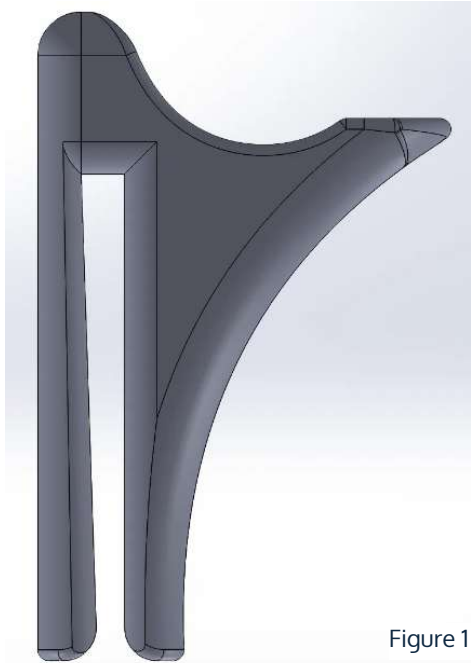


Figure 1

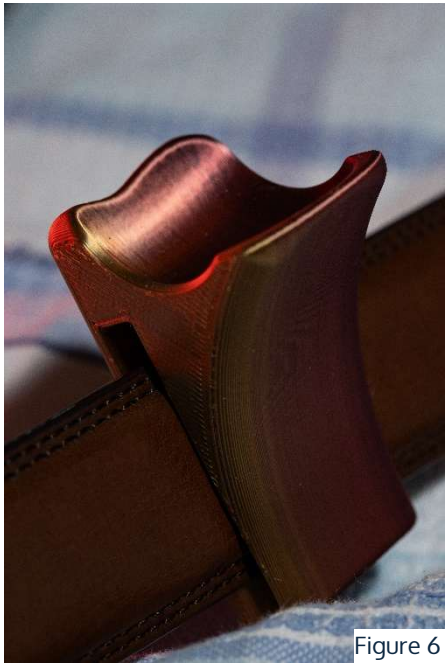


Figure 6

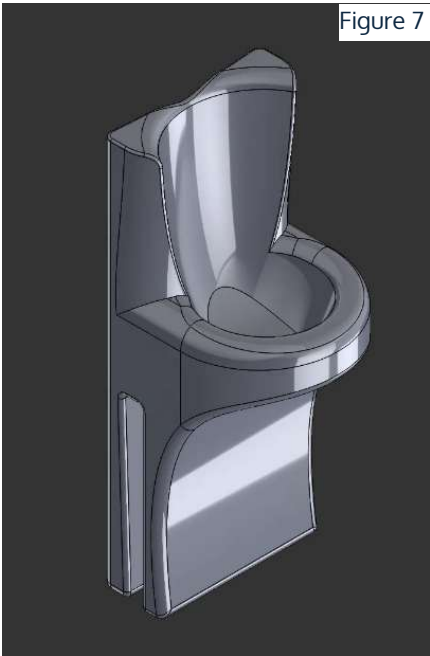


Figure 7

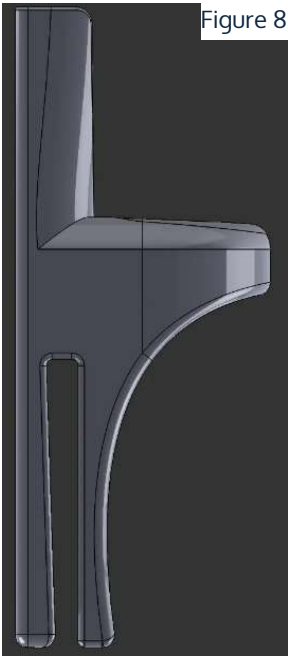
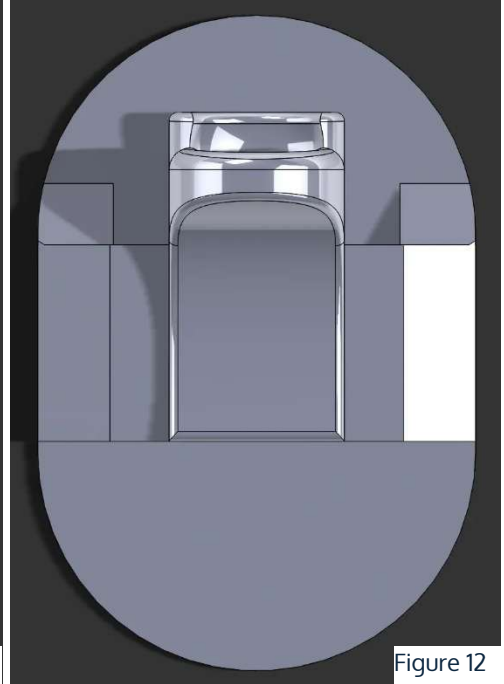
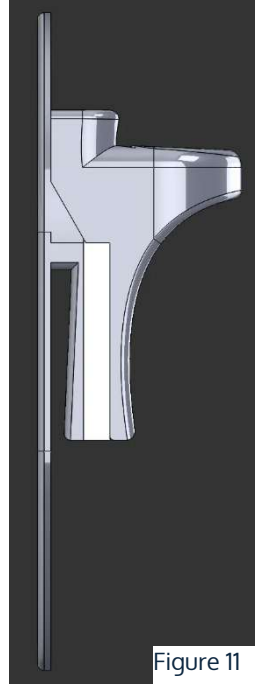
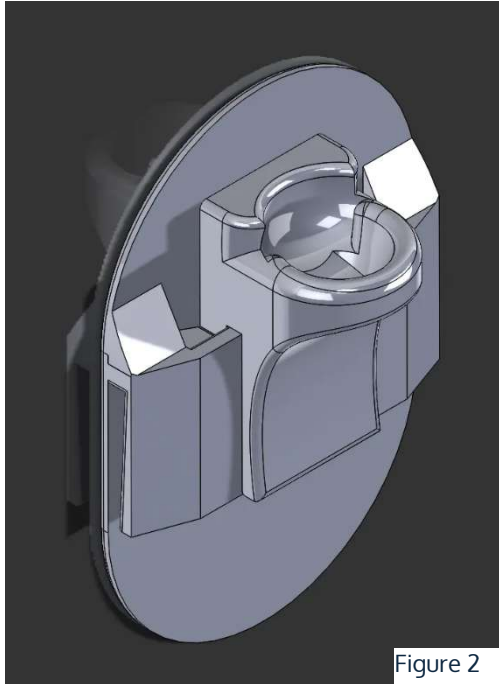


Figure 8



Figure 9

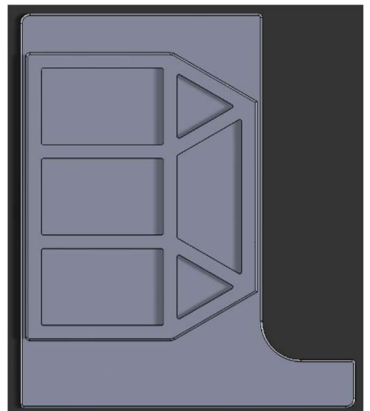
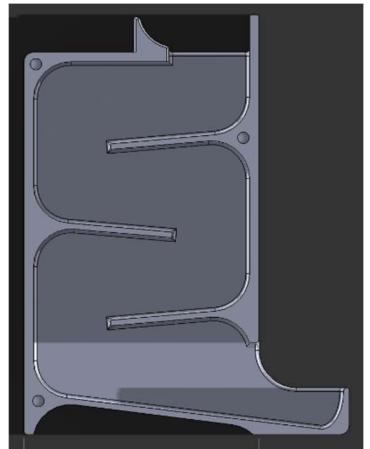


Greenhouse Juice Health Shot Dispenser

- First 3D printed project that required iterative design
- Used a partial print to test the overhang capabilities of the printer and filament, which also gave insight on the strength of print orientation and tolerances with different filaments.
 1. This also inspired a redesign which used magnets as the fastening system.
- A successful design which sparked interest in the Greenhouse Sales department and the company (the design was unveiled during an end of term co-op presentation).

Results

- There were 3 designs in the creation of this item:
 1. 1: The first did not print correctly due to geometry and the lack of an ideal print orientation (required many supports)
 2. 2: The second had a flawed connection system using a system of pins and holes. The tolerances were not designed correctly, and dimensional accuracy of the printer was noticed here.
 3. 2.5: this partial test proved unsuccessful for a rail system (the strength of the layers in this orientation was not enough, and changing the print orientation of the piece would introduce overhangs and supports) (figures 5-6)
 4. 3: The final design which used magnets for the door, improved the angles and internals of the previous designs, and was easily printable (figures 7-11)



Next Steps:

- The bottles occasionally jam going down the chute as there is wiggle room in the drop chute. Solving this may require a redesign or different tolerances.
- Find an easier way to press fit magnets, or include a different mounting system.



Figure 1

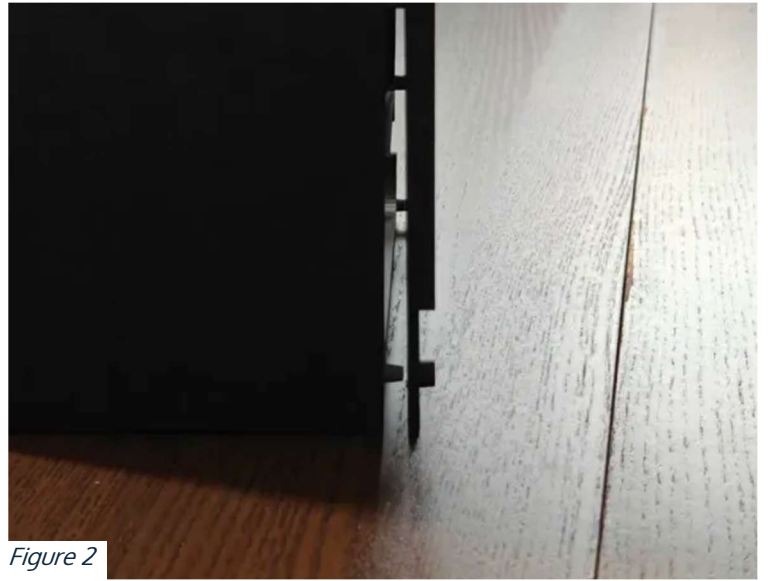


Figure 2

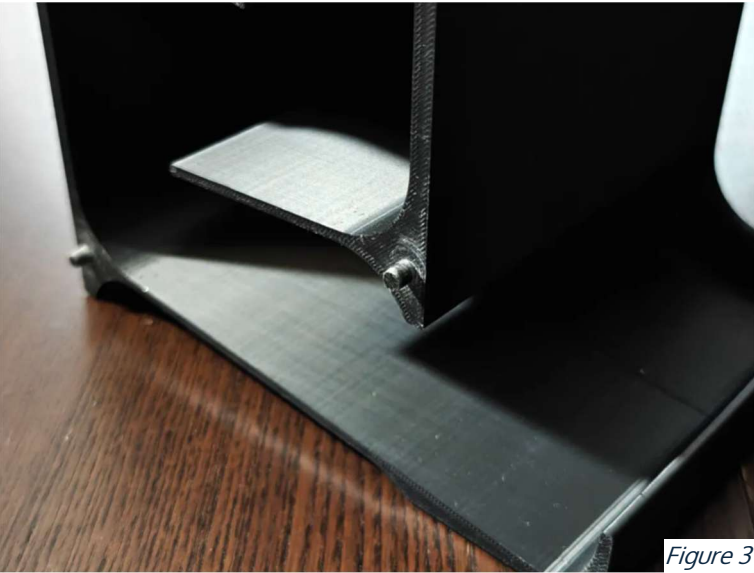


Figure 3

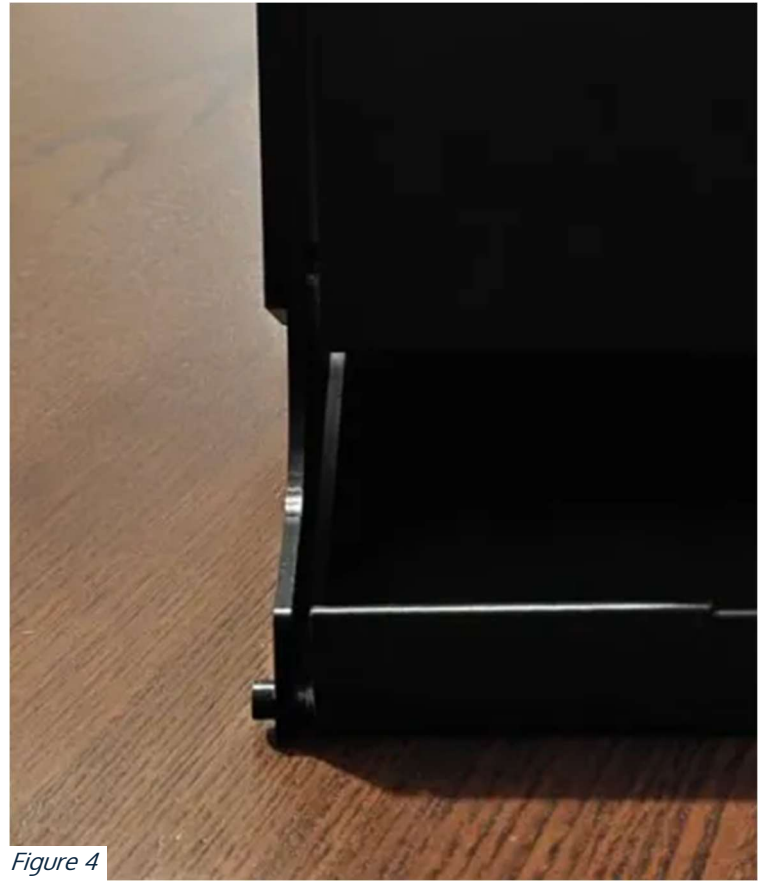


Figure 4



Figure 5



Figure 6

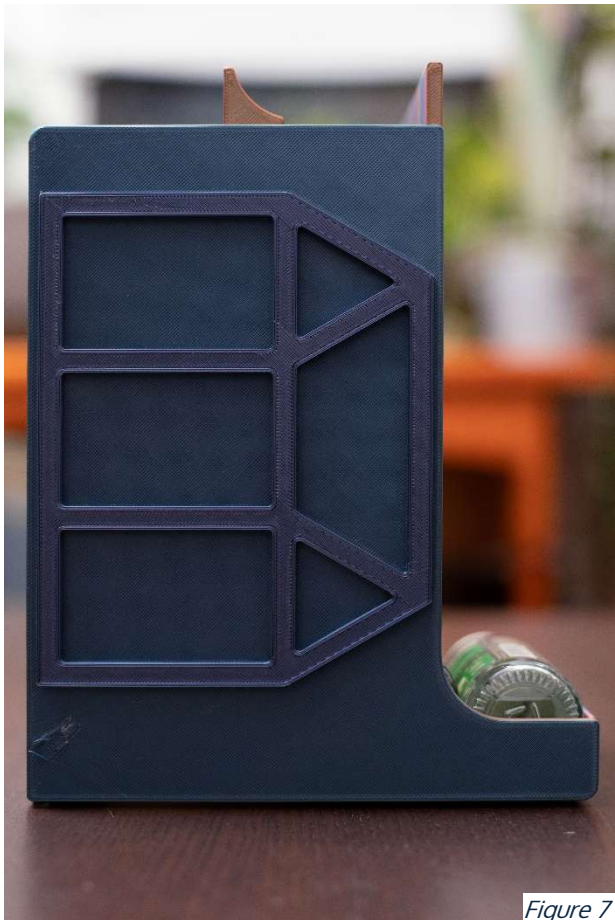


Figure 7



Figure 9



Figure 8



Figure 10

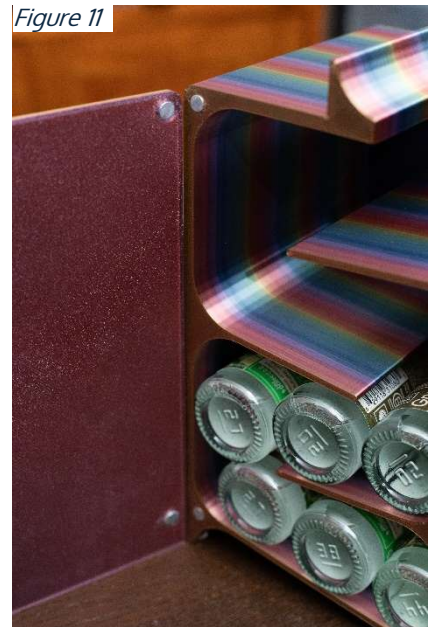


Figure 11

Personal Website

Overview

Site: <https://nathaniel2go.vercel.app/>

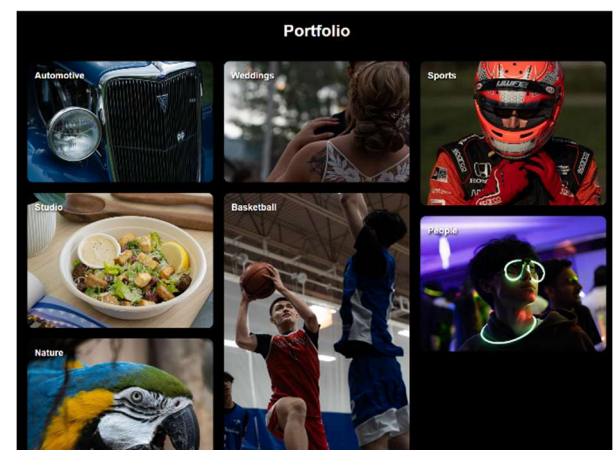
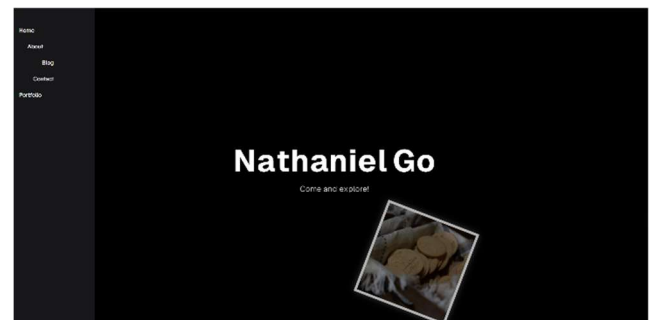
- Designed 3 versions of a personal website and portfolio:
 1. One using a template hosted locally using XAMPP and Apache
 2. Another using a heavily modified template hosted through Github Pages
 3. One built using React and hosted with Vercel
- Designed the site with best design practices and heuristics such as component placement, Fitts' law, and Universal design principles.
- Used libraries such as React, GSAP, and Tailwind for the current live version of the site

Results:

- A beautiful and reactive site used to display my photography and engineering work, as well as contact information and resume access.
- Experience working and designing the front end of a site as well as some basic wireframing

Next Steps:

- Enable a proper database, migrating existing data to tables instead of individual markdown files
- Continue to improve responsiveness and features on the site for ease of use



Automotive

[Back to Portfolio](#)

Blog

Stories, tutorials, and insights from my photography journey



May 4, 2025

Modelling a Replica Weapon - Team Fortn Two's Rocket Launcher

A comprehensive guide to building your first camera kit without breaking the bank.

[Embedded Systems](#) [CAD](#) [3D Printing](#) [3D Design](#)



April 4, 2025

Journey With Photography

Exploring the world of street photography in downtown. A journey through light, shadow, and human connection.

[Photography](#)

Animating in Blender 3.6

February 1, 2025

Animating in Blender 3.6

Understanding how to use natural light to create stunning portraits and landscapes.

[Animation](#) [3d Design](#) [Blender](#)



About Me

I'm a full time student who is passionate about learning and creating whenever the opportunity arises. Entering my second year in the University of Waterloo's Systems Design Engineering program, I am constantly approaching a mix or technical understand, attention to detail, and creativity. Whether it be in video, photography or other professional opportunities, I am excited to see more and do more.



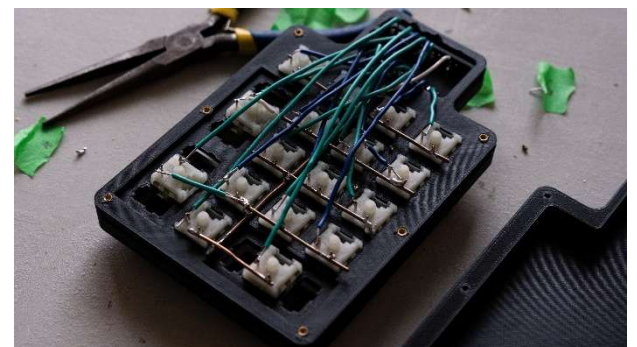
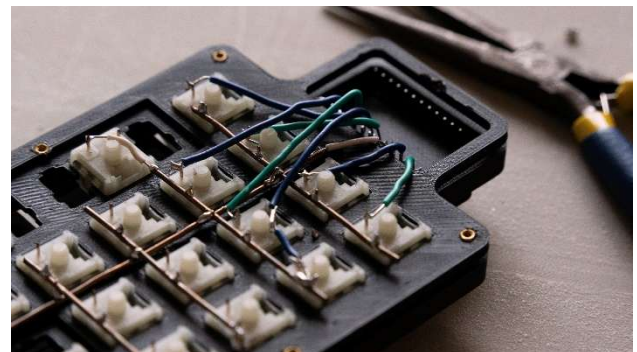
Custom 3d Printed Number Pad

Overview

- A project that combined my loved for mechanical keyboards with my first steps into the FDM and micro controller spaces.
 - Models were derived from an online model, printed on a Prusa v3 in the Waterloo Rapid Prototyping Center, using an Arduino pro micro and a skeleton system for the pin ins (to the mechanical switches)
 - The programming itself was done with a ZMK Device Tree Source and flashed using QMK.
-

Results:

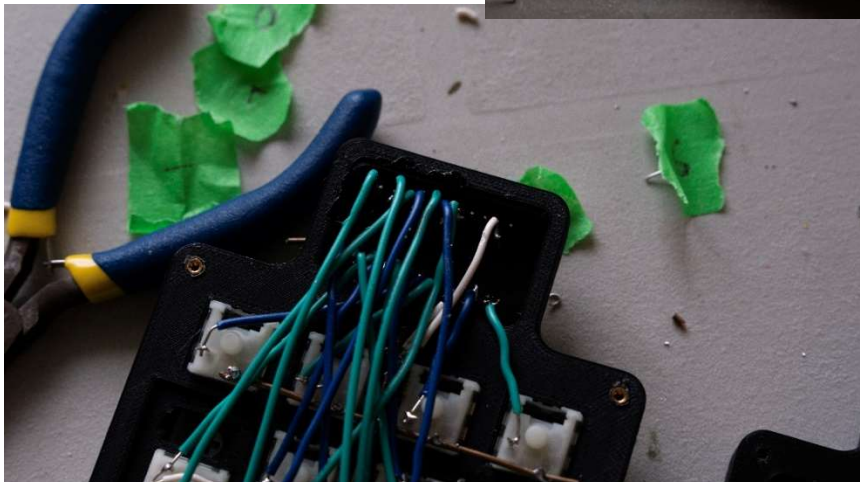
- A fully working 17 key numpad which I used for Excel work and key binds in programs like Premiere Pro, After Effects, and Blender.
- In recent months with moving, some of the joints have broken loose and have needed resoldering.
- Has exposed the benefits of using pre-manufactured metal parts instead of 3d printed fasteners (screws and inserts)



- First exposure to integrated systems and a mix of product design and hardware/software engineering concepts

Next Steps:

- Building a larger keyboard (Macro pad or stream deck type format) to use a coordinate mapping system with one way diodes, and to introduce more functionality
- Introducing more complexity into the 3d printed shell of the keyboard
- Completely modelling the keyboard from start to finish



UWFE Sponsorship Package 2026

Overview

- Link to the Sponsorship Package:
<https://static1.squarespace.com/static/59701d2159cc68dacd9e3726/t/68961d69be9bef7b7f176c81/1754668393763/2025-26+Sponsorship+Package.pdf>
- Photographed, organized, and designing the UWFE (University of Waterloo Formula Electric) sponsorship package to be sent to the team's numerous sponsors
- Primarily used Adobe Lightroom and Photoshop for photo processing, and Illustrator for design.

Contributions

- Photographed many of the photos featured in the package itself as well as other channels including the UWFE website and social media
- Fully created a design template, and fonts for the package
- Designed and laid out many of the pages featured in the final package

Results

- A professional, stylish, and complete sponsorship package highlighting the team's values and progress, the level of student support, and the reasons to sponsor the team
- The package is hosted on the live version of the UWFE website, and is also sent and distributed to sponsors via email.

